

2025 Quantitative Methods Past Questions

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UPSA SRC President (Hopeful) 26'

Question 1

(a) The demand and supply functions of a good in an industry are given by:

$$Q_d + 2P = 308$$

$$3Q_s - 0.5P = -51$$

where P , Q_d and Q_s denote the price, quantity demanded, and quantity supplied, respectively.

- (i) Determine the equilibrium price and quantity. (3 Marks)
- (ii) Suppose the government imposes an excise tax of GH¢ 26 on each good, determine the new equilibrium price and quantity. (5 Marks)
- (iii) Show how the economic agents share the tax burden. (2 Marks)

Solution

You are given two equations:

1. Demand Equation:

$$Q_d + 2P = 308$$

2. Supply Equation:

$$3Q_s - 0.5P = -51$$

Where:

- Q is quantity (the amount of the good)
- P is price
- The goal is to find the equilibrium,

where quantity demanded = quantity supplied

(i) Find Equilibrium Price and Quantity

Step 1:

At equilibrium:

$$Q_d = Q_s = Q$$

This means we set the two equations equal by solving them together.

Step 2: Solve the two equations at the same time.



We're solving:

$$1. Q + 2P = 308$$

$$2. 3Q - 0.5P = -51$$

Let's use substitution or elimination. We'll use substitution here.

From equation (1):

$$Q = 308 - 2P$$

Now plug this into equation (2):

$$3(308 - 2P) - 0.5P = -51$$

Multiply:

$$924 - 6P - 0.5P = -51$$

$$924 - 6.5P = -51$$

Now solve for P:

$$924 + 51 = 6.5P$$

$$975 = 6.5P$$

Divide both sides by 6.5

$$P = 150$$

Now plug $P = 150$ back into $Q = 308 - 2P$:

$$Q = 308 - 2(150) = 308 - 300 = 8$$

Final Answer for (i):

- Equilibrium Price (P) = GH¢ 150
- Equilibrium Quantity (Q) = 8 units

Exam Tip:

- Marks will not be awarded for solving simultaneously, so you could just use the calculator to save time.

(ii) Government Adds a Tax of GH¢26

When a tax is added, the supplier earns less, because part of the money goes to the government.

So, we need to adjust the supply equation.

Step 1: Start with the original supply equation:

$$3Q - 0.5P = -51$$

When the government puts a tax of GH¢26, the new supply equation becomes:

$$3Q - 0.5(P - 26) = -51$$

That's because suppliers now get $P - 26$ (price minus tax).

Let's simplify:

$$3Q - 0.5P + 13 = -51$$

$$3Q - 0.5P = -64$$

So the new supply equation is:

$$3Q - 0.5P = -64$$

Demand equation stays the same:

$$Q + 2P = 308$$

Step 2: Solve again like before

From the demand equation:

$$Q = 308 - 2P$$

Now substitute into the new supply equation:

$$3(308 - 2P) - 0.5P = -64$$

$$924 - 6P - 0.5P = -64$$

$$924 - 6.5P = -64$$

Now solve:

$$924 + 64 = 6.5P$$

$$988 = 6.5P$$

Divide both sides by P

$$P = 152$$

Now find Q:

$$Q = 308 - 2(152) = 308 - 304 = 4$$

Final Answer for (ii):

- New Equilibrium Price (P) = GH¢ 152
- New Equilibrium Quantity (Q) = 4 units

Exam Tip:

- When there's a tax, subtract it from P in the supply equation.
- Solve as usual after adjusting the equation.
- Be careful with signs and brackets!

(iii) How is the Tax Shared Between Consumers and Producers?

Tax per unit = GH¢ 26

Old Price = GH¢ 150

New Price = GH¢ 152

So, Consumers now pay GH¢2 more

That means:

- Consumer's share = GH¢ 2
- Producer's share = GH¢ 26 - GH¢ 2 = GH¢ 24

Final Answer for (iii):

- Consumers pay GH¢ 2 of the tax
- Producers pay GH¢ 24 of the tax

Exam Tip:

- Compare old and new price to see what the consumer pays.
- Subtract that from the tax to get the producer's share.

(b) A company specializes in the production of two distinct products: Product A and Product B. The manufacturing process for these items involves two key stages: molding and assembly. Each unit of Product A necessitates 2 hours of molding and 1 hour of assembly, while each unit of Product B requires 1 hour of molding and 3 hours of assembly. The company's manufacturing capabilities are constrained by a total of 80 hours of molding and 90 hours of assembly available per week. With a profit margin of GH¢ 50 per unit for Product A and GH¢ 60 per unit for Product B, the company's

overarching objective is to maximize its weekly profit while upholding all production and resource constraints. If x represents the number of units of Product A produced per week, and y the number of units of Product B produced per week. (i) write down the objective function and all the inequalities which must be satisfied by x and y ,

(3 Marks)

(ii) represent graphically the region which satisfies simultaneously your inequalities, (4 Marks)

(iii) how should the manufacturer arrange production to maximise total profit? (3 Marks)

TOTAL (20 MARKS)

Solution

Step 1: Understand the Problem

A company makes:

● Product A

● Product B

To make them, they use two processes (machines):

● Molding

● Assembly

Step 2: Build the Table for Time Usage

This table helps us understand how much time each product needs and how much time is available.

Time Requirement Table

● x = number of Product A made

● y = number of Product B made

ITEM	Product A	Product B	Maximum Hours
Molding	$2x$	y	80
Assembly	x	$3y$	90

Explanation:

● Product A takes 2 hours on molding and 1 hour on assembly, so we write $2x$ and x .

● Product B takes 1 hour on molding and 3 hours on assembly, so we write y and $3y$.

Step 3: Write Down the Equations (Constraints)

From the table:

1. Molding constraint:

$$2x + y \leq 80$$

2. Assembly constraint:

$$x + 3y \leq 90$$

3. Non-negativity constraint:

$$x \geq 0, y \geq 0$$

Step 4: Write the Objective Function (What to Maximize)

Profit:

● GH¢50 for each Product A (x)

● GH¢60 for each Product B (y)

So:

Objective Function:

$$\pi = 50x + 60y \text{ (}\pi \text{ means profit)}$$

Step 5: Find Intercepts for the Graph

Now we want to draw the lines for the equations.

To do this, we change the “ $\leq$ ” to “ $=$ ” and find intercepts.

For $2x + y = 80$:

● If $x = 0 \rightarrow y = 80 \rightarrow \text{Point } (0, 80)$

● If $y = 0 \rightarrow x = 40 \rightarrow \text{Point } (40, 0)$

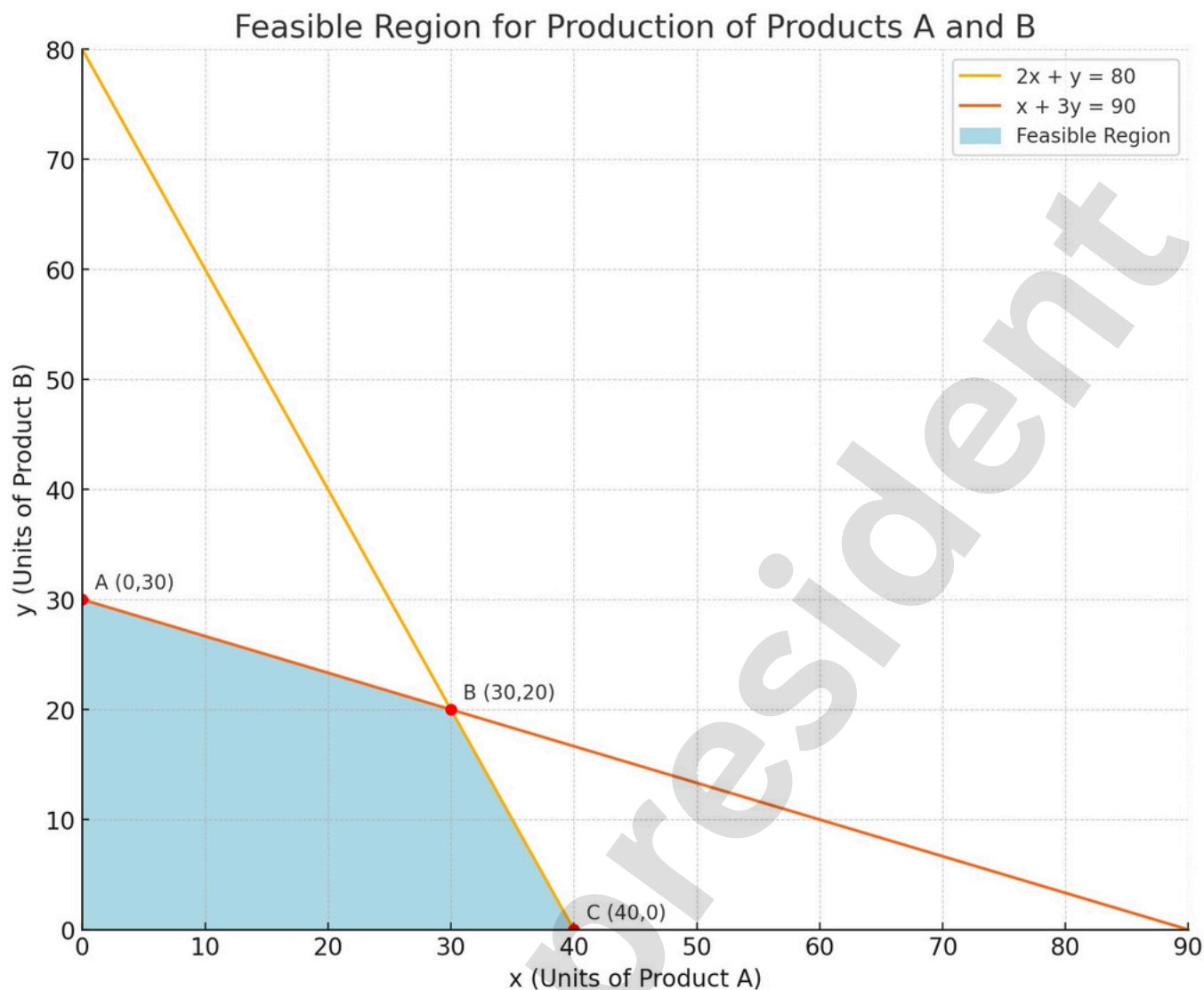
For

$x + 3y = 90$:

● If $x = 0 \rightarrow y = 30 \rightarrow \text{Point } (0, 30)$

● If $y = 0 \rightarrow x = 90 \rightarrow \text{Point } (90, 0)$

We plot these 4 points on a graph.



After drawing the graph, the corner points of the feasible region are A(0,30), B(30,20) and C(40,0). They are the set of points that satisfy all the inequalities/ give you profit but even though they all give you profit we are interested in the one that gives us the highest profit, so we solve individually for them to see which one gives us the highest profit.

So we substitute the feasible regions into the objective function ($\pi = 50x + 60y$)

Corner	Profit
A(0,30)	$\pi = 50(0) + 60(30) = 1,800$
B(30,20)	$\pi = 50(30) + 60(20) = 2,700$
C(40,0)	$\pi = 50(40) + 60(0) = 2,000$

Final Answer:

To make the maximum profit of GH¢2700 the company should:

- Make 30 units of Product A
- Make 20 units of Product B

Question 2

(a) A small retail store observes a steady increase in the number of customers over the past few months. The store manager recorded the number of customers served each month, starting from January to June, as follows:

Month	January	February	March	April	May	June
Customers	120	140	160	180	200	220

- (i) Using the formula for the n th term, find the number of customers the store served in December if the trend continues. (2 Marks)
- (ii) Find the total number of customers the store served from January to December. (2 Marks)

Solution

We can clearly see that the number of customers is increasing by 20 each month.

So:

- (b) First term (January) = $a = 120$
- (c) Common difference = $d = 20$

(i) How many customers will come in December?

December is the 12th month, so we are being asked to find the 12th term of this pattern.

Formula to use:

$$T_n = a + (n - 1)d$$

Where:

- T_n is the number of customers in the n th month,
- a is the first term (January = 120),
- d is the common difference (20),
- n is the number of the month (December = 12)

Plug in the values:

$$T_{12} = 120 + (12 - 1) \times 20 = 340$$

Answer:

The number of customers in December will be 340 if the pattern continues.

(ii) What is the total number of customers from January to December?

This is asking for the sum of the first 12 months.

Formula for sum of n terms:

$$S_n = n/2 \times [2a + (n - 1)d]$$

Where:

- S_n is the total number of customers from month 1 to month n,
- $n = 12$,
- $a = 120$,
- $d = 20$

Plug in the values:

$$S_{12} = 12/2 \times [2 \times 120 + 11 \times 20] =$$

$$6 \times [240 + 220] =$$

$$6 \times 460 = 2760$$

Answer:

The total number of customers from January to December is 2,760.

Exam Tips:

1. Always check if the pattern is an AP (ask: is the same number added each time?).
2. Know your formulas.
3. Label what each letter (a, d, n) means before you plug in — it reduces mistakes.
4. Always show your workings neatly for full marks.
5. If time allows, recheck by writing out a few terms to confirm your answer makes sense.

(d) An investor has invested in three types of bonds: Corporate bonds, Government bonds, and Municipal bonds. Let x , y , and z represent the amounts invested in thousands of dollars in each type of bond, respectively. The investor has the following constraints:

- The total investment amount is 100 thousand Ghana Cedis.

- The investor allocates three times as much money in Corporate bonds as in Municipal bonds.
 - The investment in Government bonds should be 30 thousand Ghana Cedis more than in Corporate bonds.
- (i) Generate the system of linear equations representing the above constraints. (3 Marks)
- (ii) Represent the system of equations in a matrix form $MX = N$. (1 Mark)
- (iii) Find the determinant of the matrix M. (2 Marks)
- (ii) Hence, find the amounts invested in each type of bond.

(10 Marks) TOTAL (20 MARKS)

Solution

Question Summary (In simple terms):

An investor is putting money into 3 types of bonds:

- Corporate bonds → let's call the amount invested = x (in thousands of Ghana Cedis)
- Government bonds → let's call it y
- Municipal bonds → call it z

There are 3 important rules (constraints):

1. The total investment is GH¢100,000.

$$\rightarrow \text{So: } x + y + z = 100$$

2. The amount in corporate bonds (x) is 3 times the amount in municipal bonds (z).

$$\rightarrow \text{So: } x = 3z$$

3. The amount in government bonds (y) is 30 thousand more than the corporate bonds (x).

$$\rightarrow \text{So: } y = x + 30$$

Step (i): Write the System of Linear Equations

We'll write all 3 equations based on the rules:

1. $x + y + z = 100$

2. $x = 3z$

3. $y = x + 30$

But to solve using matrices, we need the equations to look like this:

everything on one side = 0

Let's rewrite the 3 equations:

1. $x + y + z = 100 \rightarrow$ stays the same

2. $x - 3z = 0 \rightarrow$ moved $3z$ to the left

3. $-x + y = 30 \rightarrow$ moved x to the left

Step (ii): Matrix Form ($MX = N$)

Now, we'll write this in matrix form, which looks like:

$$M \times X = N$$

- M = the coefficients of x , y , and z
- $X = [x, y, z]$ (the unknowns)
- N = the answers (right-hand side of the equations)

Let's build them:

$M =$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -3 \end{bmatrix} \quad (\text{hint: there is } 0 \text{ here because there is no } Y \text{ value in the second equation})$$

$[-1 \ 1 \ 0]$ (There is 0 here as well because there is no Z value in the third equation)

X=

$[x]$

$[y]$

$[z]$

N=

$[100]$

$[0]$

$[30]$

Therefore, $MX = N$ is

$[1 \ 1 \ 1] \quad [x] \quad [100]$

$[1 \ 0 \ -3] \quad [y] = [0]$

$[-1 \ 1 \ 0] \quad [z] \quad [30]$

Step (iii): Find the Determinant of Matrix M

Matrix M is:

$[1 \ 1 \ 1]$

$[1 \ 0 \ -3]$

$[-1 \ 1 \ 0]$

Use the rule for 3×3 determinant:

$|M| =$

$$= 1 \times (0 \times 0 - (-3 \times 1)) - 1 \times (1 \times 0 - (-3 \times -1)) + 1 \times (1 \times 1 - 0 \times -1)$$

$$= 1 \times (0 + 3) - 1 \times (0 - 3) + 1 \times (1 - 0)$$

$$= 3 - (-3) + 1 = 3 + 3 + 1 = 7$$

So, Determinant = 7

Step (iv): Use Cramer's Rule to Find x, y, z

We'll find:

$$\bullet \quad x = \frac{|M_x|}{|M|}$$

$$\bullet \quad y = \frac{|M_y|}{|M|}$$

$$\bullet \quad z = \frac{|M_z|}{|M|}$$

To get $|M_x|$, replace the first column of M with N:

$M_x =$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 1 & 0 \end{bmatrix}$$

Now calculate its determinant:

$$= 100(0 \times 0 - (-3 \times 1)) - 1(0 \times 0 - (-3 \times 30)) + 1(0 \times 1 - 0 \times 30)$$

$$= 100(3) - 1(90) + 0 = 300 - 90 = 210$$

$$\text{So, } x = 210 / 7 = 30$$

Now for y: replace the second column of M with N:

$M_y =$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & 0 & 0 \end{bmatrix}$$

Use determinant rule:

$$= 1(0 \times 0 - (-3 \times 30)) - 100(1 \times 0 - (-3 \times -1)) + 1(1 \times 30 - 0 \times -1)$$

$$= 1(90) - 100(-3) + 30 = 90 + 300 + 30 = 420$$

$$\text{So } x = 420 / 7 =$$

$$= 60$$

Now for z: replace the third column of M with N:

$$M_z =$$

$$\begin{bmatrix} 1 & 1 & 100 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 30 \end{bmatrix}$$

Use determinant rule:

$$= 1(0 \times 30 - 0 \times 1) - 1(1 \times 30 - 0 \times -1) + 100(1 \times 1 - 0 \times -1)$$

$$= 1(0) - 1(30) + 100(1) = 0 - 30 + 100 = 70$$

$$\text{So, } z = 70 / 7 = 10$$

Final Answer:

- Corporate bonds (x) = GH¢ 30,000
- Government bonds (y) = GH¢ 60,000
- Municipal bonds (z) = GH¢ 10,000

Exam Tips:

1. Assign variables clearly (like x, y, z) and explain what each one means.
2. Write each sentence from the question as an equation step-by-step.

3. When asked to put in matrix form, just use the numbers in front of x, y, z.
4. Practice Cramer's Rule for solving 3 equations – it comes up a lot in matrix word problems.
5. Always double-check your determinant (one small mistake changes everything).

Question 3

A bank is interested in predicting the credit card spending of its customers based on several factors. The bank collected data from a sample of credit card holders, including their annual income, age, and credit limit. The dependent variable is the monthly credit card spending (GH¢ Y). Using a multiple

linear regression model with Annual Income (GH¢ X1), Age (X2 years), and Credit Limit (GH¢ X3) as independent variables, the following is the summary output of the fitted multiple linear regression model to the data.

<i>Regression Statistics</i>	
Multiple R	0.9926
R Square	0.9852
Adjusted R Square	0.9321
Standard Error	75.9621
Observations	7

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	1151260.71	383753.57	66.5 1	0.001
Residual	3	17310.72	5770.24		
Total	6	1168571.43			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	7.3155	3.6217	2.0199	0.1367
Annual Income (X1)	0.0188	0.0048	3.7995	0.0428
Age (X) ₂	-2.9996	6.1480	-0.4879	0.6591
Credit Limit (X3)	0.0696	0.0151	4.6093	0.0085

- (a) What is the multiple linear regression equation that represents the relationship between annual income, age, credit limit, and monthly spending? (2 Marks)
- (b) Interpret the coefficients of the independent variables of the regression equation in the context of the problem. (3 Marks)
- (c) Test, at the 0.05 level, the significance of the multiple linear regression model. (4 Marks)
- (d) State, with reasons, which of the independent variables are statistically significant. (2 Marks)
- (e) If a customer has an annual income of GH¢ 65,000, age of 40 years, and a credit limit of GH¢ 9,000, what is the predicted monthly spending based on the regression model? (2 Marks)

- (f) Find the coefficient of determination for the regression model. What does this value indicate about the goodness of fit of the model? (1 Marks)
- (g) What percentage of the total variation in sales can be attributed to other factors apart from these three? (1 Mark)
- (h) Conduct a hypothesis test to determine if Credit Limit is a statistically significant predictor of monthly spending. (4 Marks)

TOTAL (20 MARKS)

Solution

(a) Write the regression equation

Exam Tip:

Use the coefficients given in the table under “Coefficients.”

We are using this format:

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3$$

Where:

- Y = monthly spending
- a = intercept = 7.3155
- b_1 = income coefficient = 0.0182
- b_2 = age coefficient = -2.9996
- b_3 = credit limit coefficient = 0.0696

So the equation becomes:

$$\text{Monthly Spending} = 7.3155 + 0.0182X_1 - 2.9996X_2 + 0.0696X_3$$

(b) Interpret the coefficients

Exam Tip:

Talk about how much monthly spending increases or decreases when each variable increases by 1 unit.

- Income (0.0188): For every 1 extra Ghana cedi of annual income, monthly spending increases by 0.0188 GH¢, on average.
- Age (-2.9996): For every 1 extra year in age, monthly spending decreases by 2.9996 GH¢, on average.

- Credit Limit (0.0696): For every 1 extra GH¢ in credit limit, monthly spending increases by 0.0696 GH¢, on average.

(c) Test if the model is significant (0.05 level)

We're testing if the entire model works. Use the ANOVA table.

Step 1: Hypotheses

- $H_0: b_1 = b_2 = b_3 = 0$ (model is NOT useful)
- H_1 : At least one $b \neq 0$ (model is useful)

Step 2: Decision rule

- F critical ($F_{0.05, 3, 3} \approx 9.28$ (from F-table or given)
- We reject H_0 if $F > 9.28$

Step 3: Use the table

- From the table, $F = 66.51$
- Also, Significance $F = 0.001$

Step 4: Decision

- $66.51 > 9.28 \rightarrow \text{Reject } H_0$
- OR $0.001 < 0.05 \rightarrow \text{Reject } H_0$

Step 5: Interpretation

The model is significant — at least one variable is useful in predicting monthly spending.

(d) Which variables are statistically significant?

Exam Tip: Look at p-values in the last column of the table.

If it is less than 0.05 then it is significant but if it is more than 0.05 then it is not significant.

- Annual Income: $p = 0.0428 \rightarrow \text{Significant}$

- Credit Limit: $p = 0.0085 \rightarrow$ Significant

- Age: $p = 0.6591 \rightarrow$ Not significant

(e) Predict spending for a customer

Customer details:

- Income = GH¢ 65,000

- Age = 40

- Credit Limit = GH¢ 9,000

Plug into the equation:

$$Y = 7.3155 + 0.0188(65,000) - 2.9996(40) + 0.0696(9000)$$

Step-by-step:

- $0.0188 \times 65,000 = 1222$

- $-2.9996 \times 40 = -119.984$

- $0.0696 \times 9000 = 626.4$

Now add:

$$Y = 7.3155 + 1222 - 119.984 + 626.4 = 1,735.73$$

So, predicted monthly spending = GH¢ 1,735.73

(f) Find the coefficient of determination (R^2) and interpret it

From the table:

- $R^2 = 0.9852$

Interpretation:

98.52% of the variation in monthly spending is explained by the 3 variables. That means the model is very good at predicting.

(g) What % is due to other factors?

$$100\% - R^2 = 100 - 98.52 = 1.48\%$$

So, 1.48% of the variation is due to other factors not included in the model.

(h) Test if Credit Limit is a significant predictor

Now we do a hypothesis test just for Credit Limit.

Step 1: Hypotheses

- $H_0: b_3 = 0$ (Credit Limit is not useful)
- $H_1: b_3 \neq 0$ (Credit Limit is useful)

Step 2: Decision Rule

Use t-value or p-value.

- t critical (df = 3, two-tailed, 0.05) ≈ 3.182
- OR just use p-value = 0.0085

Step 3: From table

- $t = 4.6093$
- $p = 0.0085$

Step 4: Decision

- $4.6093 > 3.182 \rightarrow \text{Reject } H_0$
- OR $0.0085 < 0.05 \rightarrow \text{Reject } H_0$

Step 5: Interpretation

Credit Limit is a statistically significant predictor of monthly spending.

FINAL EXAM TIPS:

- Always state the hypotheses when doing a significance test.
- Use p-values when you're not sure of t or F critical values.
- When interpreting coefficients, always say: "On average, a 1-unit increase in X leads to ... change in Y."
- Memorize the format:

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3$$

- For prediction questions, plug the numbers one by one, use a calculator.

Question 4

(a) Evaluate the indefinite integral $\int (3x-5)^2 dx$. (2 Marks)

Solution

Step 1: What does this mean?

When you see something like $\int (3x - 5)^2 dx$, it means:

Find the function whose derivative gives $(3x - 5)^2$.

We are trying to undo differentiation — that's what integration does.

Step 2: Expand the bracket first

Before you integrate, it's often easier to remove the square by multiplying out the bracket.

Let's expand:

$$\begin{aligned}(3x - 5)^2 &= (3x - 5)(3x - 5) \\ &= 9x^2 - 30x + 25\end{aligned}$$

Step 3: Now integrate term by term

We now integrate:

$$\int (9x^2 - 30x + 25) dx$$

Let's do each part one by one using this simple rule:

$$\int x^n dx = (1 / (n + 1)) x^{n+1} + C$$

- $\int 9x^2 dx = (9 / 3)x^3 = 3x^3$

- $\int -30x dx = (-30 / 2)x^2 = -15x^2$

- $\int 25 dx = 25x$

- And don't forget: + C at the end (that's the constant of integration)

Final Answer:

$$\int (3x - 5)^2 dx = 3x^3 - 15x^2 + 25x + C$$

Exam Tip:

1. Always expand powers like $(3x - 5)^2$ before integrating. It's safer.
2. Integrate one term at a time.
3. Never forget the "+ C" at the end — even 1 mark exams will remove a mark if you forget it.
4. Double-check your signs (positive/negative) after expanding.
5. If you're unsure how to expand something, do it slowly on the side.

(b) The demand and supply functions of a good are given by $P = 300 - Q^2_D$ and

$$P = 100 + \frac{Q^2_S}{2}, \text{ where } P, Q_D \text{ and } Q_S \text{ denote price, quantity demanded and quantity}$$

supplied, respectively

- (i) Calculate the equilibrium price and quantity.
(4 Marks)
- (ii) Calculate the consumer and producer surplus at equilibrium.
(6 Marks)

Solution

Demand function: $P = 300 - Q^2$

Supply function: $P = 100 + Q^2$

We are to:

1. Find equilibrium price and quantity
2. Calculate consumer and producer surplus

(i) Find Equilibrium Price and Quantity Step 1: At equilibrium, demand price = supply price

So, set the two equations equal:

$$300 - Q^2 = 100 + Q^2$$

Now solve for Q:

$$300 - 100 = Q^2 + Q^2$$

$$200 = 2Q^2$$

$$Q^2 = 100$$

$$Q = \sqrt{100} = 10$$

Step 2: Substitute $Q = 10$ into any function to get price.

Use supply:

$$P = 100 + (10)^2 = 100 + 100 = 200$$

Final Answer:

● Equilibrium Quantity (Q) = 10 units

● Equilibrium Price (P) = 200 cedis

Exam Tip:

Always start by equating demand and supply and solve for Q. Then plug Q into either equation to get P.

(ii) Consumer and Producer Surplus (6 marks)

Consumer Surplus (CS):

CS = Area under demand curve – Actual payment

Formula:

$$CS = \int_0^{10} (300 - Q^2) dQ - (P \times Q)$$

That's:

$$CS = \int_0^{10} (300 - Q^2) dQ - (200 \times 10)$$

Step-by-Step Integration:

$$\bullet \int 300 dQ = 300Q$$

$$\bullet \int Q^2 dQ = (1/3)Q^3$$

So,

$$\int_0^{10} (300 - Q^2) dQ = [300Q - (1/3)Q^3]$$

$$= (300 \times 10 - (1/3) \times 10^3) - (0)$$

$$= (3000 - (1/3) \times 1000) =$$

$$3000 - 333.33 = 2666.67$$

Now subtract actual payment:

$$200 \times 10 = 2000$$

So:

$$CS = 2666.67 - 2000 = 666.67 \text{ cedis}$$

Producer Surplus (PS):

$$PS = (P \times Q) - \text{Area under supply curve}$$

Formula:

$$PS = (200 \times 10) - \int_0^{10} (100 + Q^2) dQ$$

Step-by-Step Integration:

$$\bullet \int 100 dQ = 100Q$$

$$\bullet \int Q^2 dQ = (1/3)Q^3$$

So:

$$\begin{aligned} \int_0^{10} (100 + Q^2) dQ &= [100Q + (1/3)Q^3] \text{ from 0 to 10} \\ &= (100 \times 10 + (1/3) \times 1000) = 1000 + 333.33 = 1333.33 \end{aligned}$$

Now subtract:

$$(200 \times 10) - 1333.33 = 2000 - 1333.33 = 666.67 \text{ cedis}$$

Final Answers:

$$\bullet \text{Consumer Surplus} = 666.67 \text{ cedis}$$

$$\bullet \text{Producer Surplus} = 666.67 \text{ cedis}$$

Exam Tip:

For surplus problems:

- Use integration to find the area under the curve.
- Always subtract the area of the rectangle ($P \times Q$) from it.

(c) "SocialBuzz," a social media platform, experiences a growth in new user registrations. Their monthly user base can be modelled by the logistic function:

$$U(t) = 10,000(1 - e^{-0.1t}),$$

where $U(t)$ is the number of registered users at time t (months since launch)

(i) How many users did SocialBuzz have at launch?

(2 Marks)

(ii) It takes approximately how many months for SocialBuzz to reach 8,000 users?

(5 Marks)

(iii) What is the market saturation level?

(1 Mark)

TOTAL (20 MARKS)

Solution

(i) How many users at launch?

Launch means $t = 0$

$$\begin{aligned} U(0) &= 10,000(1 - e^{-0.1 \times 0}) \\ &= 10,000(1 - e^0) = 10,000(1 - 1) = 0 \text{ users} \end{aligned}$$

(ii) When does $U(t) = 8000$?

We solve:

$$8000 = 10,000(1 - e^{-0.1t})$$

Step 1: Divide both sides by 10,000

$$0.8 = 1 - e^{-0.1t}$$

Step 2: Rearrange

$$e^{-0.1t} = 1 - 0.8 = 0.2$$

Step 3: Take $\ln$ (natural log) on both sides

$$-0.1t = \ln(0.2)$$

Step 4: Solve for t

$$t = \frac{\ln(0.2)}{-0.1} \approx -1.60941 / -0.1 = 16.09 \text{ months}$$

(iii) Market Saturation Level

Market saturation is the maximum number of users as time goes to infinity.

From the function:

$$\text{As } t \rightarrow \infty, e^{-0.1t} \rightarrow 0$$

$$\text{So, } U(t) = 10,000(1 - 0) = 10,000 \text{ users}$$

Final Answers for (c):

● (i) 0 users at launch

● (ii) About 16.09 months to reach 8,000 users

- (iii) Market saturation = 10,000 users

Question 5

(a) Given that $y = 24x^2 - 0.8x^3$.

(i) Find the value of x that maximizes y .
(4 Marks)

(ii) Hence, find the maximum value of y .
(1 Marks)

Solution

You are given:

$$y = 24x^2 - 0.8x^3$$

We are asked to:

1. Find the value of x that maximizes y
2. Hence, find the maximum value of y

What does this mean?

You're given a curve (a function of x).

You're being asked:

- Where is the highest point on the curve?

(This means: What value of x gives the maximum y ?)

- What is that highest value of y ?

To solve this, you must use differentiation. Differentiation tells us:

- Where the curve stops increasing and starts decreasing — that's the maximum point!

STEP 1: Differentiate y

We are given:

$$y = 24x^2 - 0.8x^3$$

Now we find the first derivative (dy/dx). This tells us the slope or rate of change.

Use this rule:

$$d/dx \text{ of } x^n = n \cdot x^{n-1}$$

Now apply it:

- $d/dx \text{ of } 24x^2 = 48x$

- $d/dx \text{ of } -0.8x^3 = -2.4x^2$

So:

$$dy/dx = 48x - 2.4x^2$$

Exam Tip 1:

Write your derivative clearly. You'll lose marks if it's wrong.

Use the power rule carefully: bring down the power, then subtract 1.

STEP 2: Set $dy/dx = 0$ (To find stationary points)

To find the maximum/minimum, we set:

$$dy/dx = 0$$

So:

$$48x - 2.4x^2 = 0$$

Factor this equation:

$$x(48 - 2.4x) = 0$$

Now solve each part:

- $x = 0$

- $48 - 2.4x = 0 \rightarrow 2.4x = 48 \rightarrow x = 20$

So we have two stationary points: $x = 0$ and $x = 20$

Exam Tip 2:

When you get a quadratic like this, always factor it. That's the easiest way to find the values of x you can also use the calculator to save time because marks will not be awarded for factorization.

STEP 3: Use second derivative to check if it's a maximum or minimum

Now we find the second derivative (d^2y/dx^2) — this tells us the shape of the curve:

- If it's positive, it's a minimum.
- If it's negative, it's a maximum.

We already had:

$$dy/dx = 48x - 2.4x^2$$

Now differentiate again to get the second derivative:

- d/dx of $48x = 48$
- d/dx of $-2.4x^2 = -4.8x$

So:

$$d^2y/dx^2 = 48 - 4.8x$$

Now plug in the x-values we found:

For $x = 0$:

$$d^2y/dx^2 = 48 - 4.8(0) = 48$$

→ Positive → So $x = 0$ is a minimum

For $x = 20$:

$$d^2y/dx^2 = 48 - 4.8(20) = 48 - 96$$

→ Negative → So $x = 20$ is a maximum

Answer for part (i):

The value of x that maximizes y is $x = 20$

Exam Tip 3:

Always show second derivative test in exams to get full marks.

STEP 4: Find the maximum value of y

We now substitute $x = 20$ into the original equation:

$$y = 24x^2 - 0.8x^3$$

Now plug in $x = 20$:

- $y = 24(20^2) - 0.8(20^3)$

● $y = 24(400) - 0.8(8000)$

● $y = 9600 - 6400$

● $y = 3200$

Answer for part (ii):

The maximum value of y is 3200 when $x = 20$

Final Answers:

(i) $x = 20$ maximizes y

(ii) Maximum value of $y = 3200$

(b) A manufacturing company produces two types of products, Product A and Product B. The joint total cost function for producing both products is given by:

$$TC = 10Q_A + Q_A Q_B + 10Q_B,$$

Where Q_A and Q_B denote the quantities of Product A and Product B respectively. If P_A and P_B

denote the corresponding prices, then the demand equations are:

$$Q_A = 50 - P_A + Q_B$$

$$Q_B = 30 + 2Q_A - P_B$$

(i) Find the maximum profit if the firm is contracted to produce a total of 15 units of either type.

(13 Marks) (ii) Hence, estimate the new optimal profit if the production quota rises by 1 unit.
(2 Marks)

TOTAL (20 MARKS)

SOLUTION:

You're told:

● The total cost function is:

$$TC = 10Q_A + Q_A Q_B + 10Q_B$$

● The demand equations are:

$$Q_A = 50 - P_A + Q_B$$

$$Q_B = 30 + 2Q_A - P_B$$

- You're to maximize profit (Revenue - Cost), given $Q_A + Q_B = 15$.

Step-by-Step Solution

Step 1: Find the Prices (P_A and P_B)

We're told how quantity demanded relates to price:

- From $Q_A = 50 - P_A + Q_B \rightarrow$

$$\text{Rearranged: } P_A = 50 - Q_A + Q_B$$

- From $Q_B = 30 + 2Q_A - P_B \rightarrow$

$$\text{Rearranged: } P_B = 30 + 2Q_A - Q_B$$

Step 2: Find Revenue

Revenue is price times quantity:

- $TRA \text{ (from A)} = P_A \times Q_A = (50 - Q_A + Q_B) \times Q_A$

- $TRB \text{ (from B)} = P_B \times Q_B = (30 + 2Q_A - Q_B) \times Q_B$

Now expand both:

- $TRA = 50Q_A - Q_A^2 + Q_AQ_B$

- $TRB = 30Q_B + 2Q_AQ_B - Q_B^2$

So Total Revenue (TR):

$$TR = 50Q_A - Q_A^2 + Q_A \cdot Q_B + 30Q_B + 2Q_A \cdot Q_B - Q_B^2$$

Now simplify:

$$TR = 50Q_A - Q_A^2 + 3Q_A \cdot Q_B + 30Q_B - Q_B^2$$

Step 3: Cost Function

You're given:

$$TC = 10Q_A + Q_A \cdot Q_B + 10Q_B$$

Step 4: Profit Function

Profit = Revenue - Cost:

$$\pi = [50Q_A - Q_A^2 + 3Q_A \cdot Q_B + 30Q_B - Q_B^2]$$

$$- [10Q_A + Q_A \cdot Q_B + 10Q_B]$$

Simplify:

$$\pi = 50Q_A - Q_A^2 + 3Q_A \cdot Q_B + 30Q_B - Q_B^2$$

$$- 10Q_A - Q_A \cdot Q_B - 10Q_B$$

$$\pi = 40Q_A - Q_A^2 + 2Q_A \cdot Q_B + 20Q_B - Q_B^2$$

Step 5: Use Constraint ($Q_A + Q_B = 15$)

Use Lagrangian Method (since there's a constraint).

Let λ be the multiplier.

$$L(Q_A, Q_B, \lambda) = \pi + \lambda(15 - Q_A - Q_B)$$

Substitute π :

$$L = 40Q_A - Q_A^2 + 2Q_A Q_B + 20Q_B - Q_B^2 + \lambda(15 - Q_A - Q_B)$$

Step 6: Take Derivatives

We take partial derivatives with respect to Q_A , Q_B , and λ .

$$\bullet \partial L / \partial Q_A = 40 - 2Q_A + 2Q_B - \lambda = 0$$

$$\bullet \partial L / \partial Q_B = 20 + 2Q_A - 2Q_B - \lambda = 0$$

$$\bullet \partial L / \partial \lambda = 15 - Q_A - Q_B = 0$$

Step 7: Solve the System

From solving the equations simultaneously, we get:

● $Q_A = 10$

● $Q_B = 5$

● $\lambda = 30$

Step 8: Find the Maximum Profit

Substitute $Q_A = 10$, $Q_B = 5$ into the profit formula:

$$\pi = 40Q_A - Q_A^2 + 2Q_A \cdot Q_B + 20Q_B - Q_B^2$$

$$\pi = 40(10) - 10^2 + 2(10)(5) + 20(5) - 5^2$$

$$\pi = 400 - 100 + 100 + 100 - 25 = 475$$

Final Answers

● (i) Maximum profit = 475

● (ii) If quota increases by 1 unit, the multiplier $\lambda = 30$ shows the profit will increase by 30.

$$\text{So new profit} = 475 + 30 = 505$$

Exam Tips:

1. Always isolate price (P) from the demand equation so you can find revenue.
2. Total Revenue = $P \times Q$ for each product.
3. Always expand and simplify carefully to avoid mistakes.
4. Use the Lagrangian method when you're given a constraint like " $Q_A + Q_B = 15$ ".
5. If they ask for the change in optimal profit from a unit increase in the constraint, the answer is just the value of λ (the Lagrange multiplier).

I wish you all the very best in your Quantitative Methods Exam, I'm with you in Prayers.

Isaac Osafo Junior (Ike)

Percentage points of the F -distribution

$\alpha = 0.05$

		Degrees of freedom for the numerator (v_1)								
		1	2	3	4	5	6	7	8	9
Degrees of freedom for the denominator (v_2)	1	161.50	199.50	215.70	224.60	230.20	234.00	236.80	238.90	240.60
	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	
		1	2	3	4	5	6	7	8	9
		Degrees of freedom for the numerator (v_1)								

$$P(F > F_{\alpha, v_1, v_2}) = \alpha$$

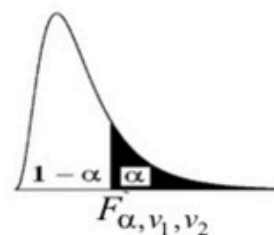
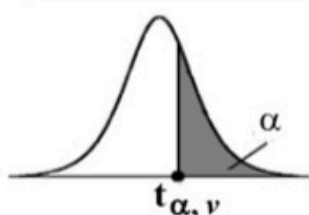


Table A.4: (continued) Percentage points of the t -distribution

v	α						
	0.025	0.05	0.10	0.15	0.20	0.30	0.40
1	12.706	6.314	3.078	1.963	1.376	0.727	0.325
2	4.303	2.920	1.886	1.386	1.061	0.617	0.289
3	3.182	2.353	1.638	1.250	0.978	0.584	0.277
4	2.776	2.132	1.533	1.190	0.941	0.569	0.271
5	2.571	2.015	1.476	1.156	0.920	0.559	0.267
6	2.447	1.943	1.440	1.134	0.906	0.553	0.265
7	2.365	1.895	1.415	1.119	0.896	0.549	0.263
8	2.306	1.860	1.397	1.108	0.889	0.546	0.262
9	2.262	1.833	1.383	1.100	0.883	0.543	0.261
10	2.228	1.812	1.372	1.093	0.879	0.542	0.260
11	2.201	1.796	1.363	1.088	0.876	0.540	0.260
12	2.179	1.782	1.356	1.083	0.873	0.539	0.259
13	2.160	1.771	1.350	1.079	0.870	0.538	0.259
14	2.145	1.761	1.345	1.076	0.868	0.537	0.258
15	2.131	1.753	1.341	1.074	0.866	0.536	0.258
16	2.120	1.746	1.337	1.071	0.865	0.535	0.258
17	2.110	1.740	1.333	1.069	0.863	0.534	0.257
18	2.101	1.734	1.330	1.067	0.862	0.534	0.257
19	2.093	1.729	1.328	1.066	0.861	0.533	0.257
20	2.086	1.725	1.325	1.064	0.860	0.533	0.257
21	2.080	1.721	1.323	1.063	0.859	0.532	0.257
22	2.074	1.717	1.321	1.061	0.858	0.532	0.256
23	2.069	1.714	1.319	1.060	0.858	0.532	0.256
24	2.064	1.711	1.318	1.059	0.857	0.531	0.256
25	2.060	1.708	1.316	1.058	0.856	0.531	0.256
26	2.056	1.706	1.315	1.058	0.856	0.531	0.256
27	2.052	1.703	1.314	1.057	0.855	0.531	0.256
28	2.048	1.701	1.313	1.056	0.855	0.530	0.256
29	2.045	1.699	1.311	1.055	0.854	0.530	0.256
30	2.042	1.697	1.310	1.055	0.854	0.530	0.256
40	2.021	1.684	1.303	1.050	0.851	0.529	0.255
60	2.000	1.671	1.296	1.045	0.848	0.527	0.254
120	1.980	1.658	1.289	1.041	0.845	0.526	0.254
∞	1.960	1.645	1.282	1.036	0.842	0.524	0.253
v	0.025	0.05	0.10	0.15	0.20	0.30	0.40
α							



$$P(T > t_{\alpha, v}) = \alpha$$

$$1. \quad \frac{dy}{dx} = nx^{n-1}$$

$$2. \quad CS = \int_0^{Q_0} (\text{demand function}) dQ - P_0 Q_0$$

$$PS = P_0 Q_0 - \int_0^{Q_0} (\text{supply function}) dQ$$

$$3. \quad \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}.$$

$$|\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}.$$

$$4. \quad \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} \det(\mathbf{A}) &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{32}a_{21} - a_{22}a_{31}). \end{aligned}$$

$$5. \quad TR = PQ, \quad \pi = TR - TC.$$

$$6. \quad P_t = P_0(1+i)^t$$

$$7. \quad \frac{d(TC)}{dQ} = MC.$$

$$8. \quad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

$$9. \quad T_n = a + (n-1)d.$$

$$10. \quad S_n = \frac{n}{2} [2a + (n-1)d].$$

$$11. \quad T_n = ar^{n-1}$$

$$12. \quad S_n = \frac{a(1-r^n)}{1-r}, \quad r \neq 1.$$